QALA

Solution Factory Operating System

Extended Workflows & Use Cases — Industry Edition v1.0

Field	Value
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System	Qala Solution Factory OS
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1 Introduction & Purpose

This document describes how users across every industry and persona type leverage the Qala Solution Factory Operating System to conceive, build, deliver, and operate solutions — from a hobbyist's first mobile app to an enterprise agricultural management platform.

Qala is industry-agnostic by design. Its structured solution model, managed SDEs, hermetic build system, and AI-driven intelligence apply equally to a fashion startup launching a DTC e-commerce platform, a government agency modernising citizen services, a musician self-publishing an independent album store, and a research lab managing computational experiment pipelines.

Each industry section in this document follows a consistent pattern:

- Industry context — who the users are and what problems they are solving.
- Solution ideas — the types of apps, products, services, goods, and platforms they create.
- Use cases — step-by-step workflows showing how Qala supports their specific journey.
- Qala platform integration — which Qala services, SDEs, pipelines, and AI capabilities are engaged.

> *All workflows in this document are built on the Qala platform primitives defined in the Qala SDD v2.0 and the Qala Workflows & Use Cases v1.0. This document extends those with industry-specific context, actor personas, and solution examples.*

1.1 Persona Types Covered

Persona	Description
App Idea Holder	An individual or small team with a software application concept, seeking to build and launch it.
AI App Builder	A developer or team building AI-powered products: LLM wrappers, computer vision tools, recommendation engines.
Product Creator	Someone designing a physical or digital product for commercial distribution.
CPG Maker	A consumer packaged goods producer managing formulation, production, packaging, and distribution.
Service Business	A business delivering ongoing services: consulting, coaching, SaaS, managed services.
Platform Builder	A team building a marketplace, API platform, or multi-sided business platform.
Enterprise Operator	A large organisation managing many solutions, teams, and environments at scale.
Independent Professional	A freelancer, consultant, hobbyist, or enthusiast building solutions for themselves or clients.
Researcher / Academic	A research lab or academic institution managing experiments, datasets, and publications.
Government / Public Sector	A government agency building or managing citizen-facing and internal solutions.

2 App Ideas & Software Product Builders

💡 App Idea Holders & Software Product Builders

From napkin sketch to production app — Qala manages the full journey.

These users have a software concept — a mobile app, web platform, SaaS tool, or API product — and need a structured environment to go from idea to live product. They may be solo developers, small founding teams, or product studios.

UC-APP-01 — Validate a Software App Idea

Actor: Idea Holder (non-technical or technical)

Trigger: User has a software app concept and wants to validate it before investing in development.

Goal: A prototype SDE is provisioned; a working clickable prototype or MVP is built and validated with target users.

Workflow

Idea Entry	User describes their app idea in the Qala portal: name, category, target audience, core problem, and key features.
AI Analysis	AI Agent analyses the idea: identifies solution type (Application), suggests initial architecture, flags potential technical risks, and recommends comparable solutions in the marketplace.
Prototype SDE	Qala provisions a time-boxed Prototype SDE with a full-stack template matching the described stack (e.g., React + Node + Postgres).
Rapid Build	Developer builds a clickable prototype or thin MVP within the sandbox. No production infrastructure costs incurred.
User Testing	Prototype is shared with test users. Feedback collected via integrated feedback forms. AI analyses responses and generates a structured insight report.
Decision Gate	User reviews AI insight report and decides: Build Full Solution, Pivot (return to prototype), or Archive.
Promote	On "Build" decision: prototype SDE is promoted to a full developer SDE. Solution record created. CI/CD pipeline configured.

Qala Platform Integration

- Prototype SDE — sandboxed, time-bounded, resource-capped.
- AI Agent — idea analysis, architecture suggestions, feedback analysis.
- Solution Management — solution record linked to prototype outcome.
- Test Management — user testing sessions tracked and linked to feature records.

UC-APP-02 — Build and Launch a SaaS Product

Actor: Founding Team (Developer + PM)

Trigger: Team decides to build a commercial SaaS product after idea validation.
Goal: SaaS product is built, tested, benchmarked, released, and live for paying customers.

Workflow

Solution Created	POST /solutions with type: Application. SDE configured with SaaS stack. CI/CD pipeline activated.
Feature Development	Development follows UC-DEV-01. Each feature branch triggers hermetic CI: build, SAST, unit tests, SCA.
Billing & Auth Features	Payment integration (Stripe etc.) flagged as security-sensitive. SE conducts threat model for payment flows. DAST run against payment endpoints.
Beta Release	Release created with type: Minor. Gates evaluated. Deployment to staging via Blue/Green. Beta users invited.
Performance Benchmarking	Benchmarking Service runs load tests. AI Agent flags p99 latency regression. Developer optimises query.
Public Launch	All release gates pass. Production deployment. Monitoring active. AI anomaly detection watching error rates.
Ongoing Ops	Continuous CI/CD for updates. SEM monitoring. AI generating optimisation recommendations weekly.

Qala Platform Integration

- Hermetic Build + CI/CD — every release is attested and gated.
- Release Management — all 8 gates enforced before production.
- SEM — threat model for payment flows; DAST on payment endpoints.
- Benchmarking — performance validated before each major release.
- AI Agent — continuous optimisation recommendations post-launch.

3 AI App Builders

🤖 AI Application Builders

LLM wrappers, computer vision tools, recommendation engines, and intelligent agents — built with Qala's AI-native workflows.

AI app builders are developing applications powered by machine learning models — LLM-based chatbots, image recognition services, recommendation engines, autonomous agents, or AI-powered analytics products. These solutions have unique lifecycle needs: model versioning, prompt management, evaluation pipelines, and continuous model performance monitoring.

UC-AI-APP-01 — Build and Version an LLM-Powered Application

Actor: AI Developer, ML Engineer

Trigger: Team is building a product where the core value is delivered by a large language model.

Goal: LLM application is built with full model version tracking, prompt management, evaluation pipelines, and safe production deployment.

Workflow

SDE Setup	AI-optimised SDE provisioned: Python runtime, model inference libraries (PyTorch/HuggingFace), vector DB client, API framework. All pinned to exact versions in hermetic environment.
Model Registry	External model (e.g., GPT-4o, Claude Sonnet, Llama 3) referenced in artefact registry with version, provider, digest, and API contract version.
Prompt Management	Prompts stored as versioned artefacts in the Workspace/CMS. Changes to prompts trigger CI evaluation pipeline.
Evaluation Pipeline	CI pipeline includes: automated evaluation suite (correctness, safety, latency, cost per token). Regressions in eval scores fail the build.
Safety Gate	SEM runs content safety checks on evaluation outputs. Harmful outputs trigger security review before promotion.
A/B Model Testing	Two model versions deployed via Canary strategy. Evaluation metrics compared in real traffic. Winning version promoted to 100%.
Cost Monitoring	Data Platform tracks token usage, cost per request, and latency. AI Agent alerts when cost per request exceeds threshold.

Qala Platform Integration

- Artefact Registry — model versions tracked with provider, version, and API contract.
- Workspace/CMS — prompts versioned alongside code.
- CI/CD — evaluation pipeline as a first-class CI stage.
- SEM — content safety checks before production promotion.
- Benchmarking — cost, latency, and quality benchmarks per model version.
- Canary Deployment — safe A/B model rollout.

UC-AI-APP-02 — Manage an AI Computer Vision Product

Actor: ML Engineer, Product Manager
Trigger: Team is building an image recognition or object detection service.
Goal: Model is trained, evaluated, versioned, deployed, and monitored with drift detection.

Workflow

Dataset Management	Training datasets stored in SDE Workspace with version tags and provenance. Data pipeline runs preprocessing and augmentation.
Training Pipeline	Training job defined as a CI/CD pipeline stage. Output model artefact stored in registry with training metadata: dataset version, hyperparameters, evaluation metrics (mAP, precision, recall).
Model Evaluation	Evaluation test suite runs against held-out test set. Performance gates: mAP must exceed baseline by ≥2%, inference latency ≤50ms.
Container Build	Model packaged into inference container. Container image scanned by SEM. Signed and attested.
Canary Deployment	New model version deployed to 10% of inference traffic. Accuracy and latency monitored against live distribution.
Drift Detection	AI Agent monitors prediction confidence distribution. Sudden confidence drop triggers alert: possible data drift or adversarial input.
Retraining Trigger	Data drift alert triggers new training pipeline run with updated dataset. Model lifecycle continues.

Qala Platform Integration

- Data Platform — dataset versioning, preprocessing pipelines, and drift detection.
- Artefact Registry — model versions with full training provenance.
- Hermetic Build — training jobs run in reproducible containerised environments.
- SEM — container image scanning for inference service.
- AI Agent — prediction drift monitoring and retraining triggers.

4 Fashion & Apparel

👗 Fashion & Apparel

From independent designers to global brands — managing collections, DTC e-commerce, supply chains, and trend analytics.

Fashion companies use Qala to build and operate: DTC e-commerce platforms, collection management systems, supply chain tracking tools, virtual try-on applications, trend analytics services, and sustainable sourcing platforms. Both indie designers and large fashion houses find applicable workflows.

UC-FASHION-01 — Launch a DTC Fashion E-Commerce Platform

Actor: Fashion Entrepreneur, Developer

Trigger: Independent designer or fashion brand wants to sell directly to consumers online.

Goal: A DTC e-commerce platform is live: product catalogue, cart, checkout, inventory, and fulfilment integration.

Workflow

Solution Design	Solution type: Platform. SDE provisioned with e-commerce stack. Product catalogue schema defined using Qala's solution structure model: System (Store) > Application (Catalogue, Checkout, Inventory).
Catalogue Build	Product catalogue application built: SKU management, size/colour variants, image assets stored in Workspace/CMS, pricing engine.
Payment & Security	Payment integration threat-modelled by SEM. PCI-DSS considerations documented. DAST run against checkout flows. Security gate required before payment feature goes live.
Inventory & Fulfilment	Inventory management application integrated with fulfilment partner API. State messages track stock levels; event messages trigger reorder workflows.
Launch	Release created. All gates pass. Blue/Green deployment to production. Performance benchmarks confirm checkout latency < 2s under load.
Season Cycle	Each collection launch follows the release management workflow. AI Agent tracks conversion rates per collection and surfaces optimisation suggestions.

Qala Platform Integration

- Solution Structure Model — store modelled as System with Catalogue, Checkout, Inventory applications.
- SEM + DAST — PCI-DSS-aligned payment flow security.
- Release Management — collection launches governed by release gates.
- AI Agent — conversion analytics and merchandising recommendations.
- Benchmarking — checkout performance validated per release.

UC-FASHION-02 — Build a Virtual Try-On AI Feature

Actor: ML Engineer, Fashion Brand Tech Team
Trigger: Brand wants to add an AI-powered virtual try-on experience to their e-commerce platform.
Goal: Virtual try-on service is live, integrated into the product catalogue, and monitored for quality.

Workflow

Prototype	Prototype SDE provisioned. Computer vision model (garment segmentation + body pose estimation) evaluated on sample dataset.
Model Pipeline	Training pipeline configured per UC-AI-APP-02. Try-on model artefact versioned in registry.
Integration	Try-on service deployed as a new Application within the Store System. Interface contract defined: accepts product SKU + user image, returns composite image.
UX Testing	Test plan created for UX quality: user panel rates realism and usability. AI analyses feedback scores.
Canary	Try-on feature launched via feature flag to 10% of users. Engagement and add-to-cart rates monitored.
Full Rollout	Metrics positive. Feature flag opened to 100%. AI Agent tracks ongoing usage and quality drift.

Qala Platform Integration

- Prototype SDE — model evaluation before full build commitment.
- AI Model Registry — try-on model versioned with training dataset and metrics.
- Solution Structure Model — try-on service as Application within the Store System.
- Feature Flags + Canary — risk-managed rollout.

5 Technology & Software

💻 Technology & Software Companies

ISVs, dev tools, API platforms, SaaS products, and developer infrastructure — the home turf of Qala.

Technology companies are the most natural Qala users. They build developer tools, API products, enterprise SaaS platforms, infrastructure software, and open-source projects. Qala provides the full factory infrastructure that these teams previously cobbled together from disparate tools.

UC-TECH-01 — Operate a Multi-Product Software Company on Qala

Actor: CTO, Engineering Manager, Developer Teams

Trigger: A software company with multiple products needs a unified, governed development factory.

Goal: All products are developed, released, and operated from a single Qala Solution Factory with consistent standards.

Workflow

Factory Setup	Solution Factory created. Each product becomes a System within the factory. Shared SDEs provisioned for common libraries. Factory-wide coding standards, security policies, and toolchain templates defined.
Per-Product SDEs	Each product team gets dedicated SDEs. SDEs enrolled in the factory. Factory policies automatically applied.
Shared Artefact Registry	Common libraries published to shared artefact registry. Dependency management ensures all products reference approved, scanned versions.
Cross-Product Benchmarking	Benchmarking Service produces weekly quality report comparing all products: defect density, test coverage, deployment frequency, lead time.
Platform Governance	PA enforces: no direct production deployments without approved CCR; all builds must be hermetically attested; Critical CVEs resolved within 24h SLA.
AI Insights	AI Agent generates cross-product recommendations: "Product B's authentication module has 3× lower defect density than Product A — suggest adoption of its patterns".

Qala Platform Integration

- Solution Factory — all products governed under a single factory.
- Cross-product Benchmarking — unified quality visibility.
- Shared Artefact Registry — consistent dependency governance.
- AI Agent — cross-product pattern learning and recommendations.
- Change Control + SEM — consistent governance across all products.

UC-TECH-02 — Build and Operate a Public Developer API Platform

Actor: Platform Engineering Team, API Product Manager

Trigger: Company wants to expose its capabilities as a public API product.
Goal: API platform is live with versioned endpoints, developer portal, rate limiting, SLA monitoring, and SDK generation.

Workflow

API Design	API contract defined as OpenAPI spec, stored in Workspace/CMS as a versioned artefact. Changes to spec trigger contract-testing CI pipeline.
Versioning Strategy	API versions managed as solution versions. Each major version is a separate Application within the Platform System. Breaking changes require CCR.
SDK Generation	SDK generation toolchain runs automatically on API spec changes. SDKs (Python, TypeScript, Go, Java) published as versioned artefacts.
Developer Portal	Developer portal Application deployed with: interactive API docs, auth management, usage dashboards, and changelog.
SLA Monitoring	Benchmarking Service tracks API p99 latency and uptime. SLA breaches trigger alerts and incident records.
Deprecation Workflow	Deprecated API versions follow change control: deprecation notice published, sunset date set, migration guide created, finally version decommissioned via SDE archival workflow.

Qala Platform Integration

- Workspace/CMS — OpenAPI spec as a versioned artefact.
- Toolchain — automated SDK generation on spec change.
- Release Management — API version promotions governed by release gates.
- Benchmarking — SLA monitoring and latency tracking.
- Change Control — breaking API changes require approved CCRs.

6 Agriculture & Agribusiness

🌾 Agriculture & Agribusiness

Precision farming platforms, supply chain traceability, crop analytics, and rural marketplace solutions.

Agricultural businesses use Qala to build and operate: precision farming dashboards, IoT sensor data platforms, crop yield prediction services, supply chain traceability systems, commodities trading tools, and rural e-commerce platforms. Solutions span the range from individual farm management apps to enterprise agribusiness platforms.

UC-AGRI-01 — Build a Precision Farming IoT Platform

Actor: AgTech Developer, Farm Operations Manager

Trigger: Agricultural company wants to unify sensor data from across their farms into an actionable analytics platform.

Goal: IoT data platform is live: ingesting sensor data, running analytics, generating yield predictions, and surfacing alerts.

Workflow

Solution Model	Solution type: Platform. System: Farm Intelligence Platform. Applications: Sensor Ingestion, Crop Analytics, Alert Management, Farmer Dashboard.
Data Pipeline	Data Platform configured: IoT sensor messages (soil moisture, temperature, humidity, GPS) ingested via event stream. Message type: Event (dynamic). Data structures define sensor payload schema.
Edge SDE	Lightweight SDE deployed to on-farm edge hardware. Hermetic build ensures consistent firmware across all farm devices.
Predictive Analytics	AI Agent trained on historical yield and weather data. Crop yield predictions published to AI_RECOMMENDATIONS. Farmer dashboard surfaces irrigation and harvest timing recommendations.
Alert Management	SEM-style alert system configured for agronomic thresholds: soil moisture below 20% triggers irrigation alert; frost probability >70% triggers crop protection alert.
Traceability	Each crop batch assigned a provenance record in the artefact registry (repurposed for produce batches): batch ID, field ID, inputs used, harvest date, chain-of-custody.

Qala Platform Integration

- Solution Structure Model — multi-application Platform with well-defined interfaces.
- Data Platform — IoT event ingestion and analytics pipelines.
- Edge SDE — hermetic builds for on-farm hardware consistency.
- AI Agent — yield prediction and agronomic recommendations.
- Artefact Registry — produce batch traceability records.

UC-AGRI-02 — Launch a Farm-to-Consumer Marketplace

Actor: Agribusiness Product Team

Trigger: Agricultural cooperative wants to sell directly to consumers and restaurants.

Goal: Marketplace platform live with producer onboarding, inventory, ordering, and delivery integration.

Workflow

Platform Design	Solution type: Platform. Applications: Producer Portal, Consumer Marketplace, Order Management, Delivery Integration.
Producer Onboarding	Each farm producer onboarded as a data entity. Produce listings managed with availability (state messages) and order events (event messages).
Cold Chain Tracking	Delivery Application tracks temperature logs for cold chain compliance. Alerts fire if temperature threshold exceeded in transit.
Seasonal Releases	Each seasonal produce cycle treated as a minor release. New produce categories and pricing rules deployed via change control.
Quality Benchmarking	Consumer rating data fed into Analytics. AI Agent surfaces producer quality rankings and flags declining quality trends.

Qala Platform Integration

- Solution Structure — multi-sided Platform with Producer and Consumer applications.
- Event/State Messages — order events and availability state tracked through typed interfaces.
- Change Control — seasonal catalogue updates governed.
- AI Agent — producer quality analytics and trend detection.

7 Delivery, Shipping & Logistics

🚚 Delivery, Shipping & Logistics

Route optimisation, last-mile delivery apps, freight management, and real-time tracking platforms.

Logistics companies build: real-time shipment tracking platforms, route optimisation services, warehouse management systems, driver apps, freight marketplace platforms, and customs documentation tools. These solutions require high reliability, real-time data, and complex integrations with carriers and regulatory systems.

UC-LOGISTICS-01 — Build a Real-Time Shipment Tracking Platform

Actor: Logistics Platform Developer, Operations Manager

Trigger: Logistics company wants to provide customers and internal teams with real-time shipment visibility.

Goal: Tracking platform live: events ingested from carriers, status updated in real time, customers notified proactively.

Workflow

Solution Model	Platform System. Applications: Carrier Integration (ingests events from carrier APIs), Tracking Engine (processes and stores shipment state), Customer Portal (displays tracking), Notification Service (proactive alerts).
Event Contracts	Shipment events typed as Event (dynamic) messages: SHIPMENT_CREATED, IN_TRANSIT, OUT_FOR_DELIVERY, DELIVERED, EXCEPTION. Schemas stored in Workspace/CMS.
Carrier SDEs	Each carrier integration deployed in its own SDE for isolation. SDE factory enforces shared event contract standards.
SLA Benchmarking	Benchmarking Service tracks: event ingestion latency (target <5s from carrier), update-to-customer notification time (target <30s).
Reliability	Hermetic build ensures carrier integration SDEs are reproducible. Rollback capability means any broken integration can revert to last clean snapshot within 5 minutes.
AI Prediction	AI Agent predicts delivery delays based on carrier performance patterns, weather data, and route congestion. Proactive notifications sent to customers before SLA breach.

Qala Platform Integration

- Solution Factory — carrier integrations governed as factory members.
- Event Architecture — shipment lifecycle modelled as typed event messages.
- Benchmarking — SLA tracking for ingestion and notification latency.
- SDE Rollback — fast recovery from carrier API changes.
- AI Agent — delay prediction and proactive customer notifications.

UC-LOGISTICS-02 — Build and Operate a Route Optimisation Service

Actor: Logistics Tech Team, Fleet Manager
Trigger: Delivery company wants to reduce delivery costs and time via AI-driven route optimisation.
Goal: Route optimisation service is live, integrated with driver app, and continuously improving.

Workflow

Prototype	Prototype SDE built with optimisation algorithm (VRP solver). Evaluated against historical delivery data: cost reduction vs. baseline measured.
Production Build	Service deployed as Application within logistics Platform. Interface: accepts orders + driver availability, exports optimised route plan.
Driver App Integration	Driver mobile app (separate Application) consumes route plan via typed interface. Route updates delivered as event messages.
Performance Gate	Benchmark gate: optimisation must reduce average route distance by ≥8% vs. manual planning. Gate evaluated on each release.
Continuous Improvement	AI Agent analyses completed route data vs. planned routes. Identifies systematic deviations and suggests model retraining.

Qala Platform Integration

- Prototype SDE — algorithm evaluation before production commitment.
- Interface Contracts — typed interfaces between optimisation service and driver app.
- Benchmark Gate — quantified improvement required for every release.
- AI Agent — continuous model improvement from operational data.

8 Manufacturing

🏭 Manufacturing

Production line management, quality control systems, equipment monitoring, supply chain platforms, and digital twin applications.

Manufacturing companies use Qala to build: production management systems (MES), equipment health monitoring (predictive maintenance), quality control automation, supply chain visibility platforms, digital twin applications, and worker safety systems. Solutions must be highly reliable and often integrate with physical hardware.

UC-MFG-01 — Build a Predictive Maintenance Platform

Actor: Manufacturing Engineer, IoT Developer

Trigger: Factory wants to predict equipment failures before they cause unplanned downtime.

Goal: Predictive maintenance service is live: sensor data ingested, failure predictions generated, maintenance work orders created.

Workflow

Solution Model	System: Factory Intelligence Platform. Applications: Sensor Ingestion, Anomaly Detection, Maintenance Scheduling, ERP Integration.
Data Pipeline	Machine sensor data (vibration, temperature, pressure, current draw) ingested as event messages. Data Platform runs feature engineering pipeline.
ML Model	Failure prediction model trained on historical maintenance records. Model artefact versioned in registry with: training dataset version, feature set, evaluation metrics (precision/recall on failure prediction).
Hermetic Edge SDE	Edge SDE deployed to factory floor hardware. All sensor collection and inference runs locally for low latency and resilience to network outages.
Work Order Integration	Predicted failure events trigger automated work order creation in ERP system via typed interface export.
Performance Benchmarking	Benchmarking: model evaluated monthly on actual vs. predicted failures. Precision and recall tracked over time. AI Agent alerts on model drift.

Qala Platform Integration

- Data Platform — sensor ingestion and feature engineering pipelines.
- ML Model Registry — failure prediction model with full training provenance.
- Hermetic Edge SDE — reproducible edge deployment for factory floor.
- Interface Contracts — typed event exports to ERP integration.
- AI Agent — model drift detection and retraining triggers.

UC-MFG-02 — Manage a Digital Product Lifecycle (PLM) on Qala

Actor: Product Engineer, Manufacturing PM
Trigger: Manufacturer needs to manage product design, BOM, manufacturing specs, and revision history.
Goal: Digital PLM system manages all product data from design to end-of-life with full version control and change governance.

Workflow

Product as Solution	Each manufactured product modelled as a Solution in Qala: type Good. System contains Applications for: CAD Asset Management, BOM Management, Manufacturing Spec, Quality Standard.
Version Control	All design files, BOMs, and specs stored in SDE Workspace with full version history. Engineers check out, modify, and commit changes.
Change Control	Any change to a production BOM or manufacturing spec requires a CCR. CCB includes: Product Engineer, Quality Manager, Manufacturing Manager. High-risk changes require physical sample approval before activation.
Release to Manufacturing	Product release follows the release management workflow. Release record includes: BOM version, manufacturing spec version, quality acceptance criteria.
End of Life	Product decommission follows SDE archive workflow. All historical records retained per retention policy.

Qala Platform Integration

- Solution Model — manufactured goods modelled as Solutions of type Good.
- Version Control — all product data versioned.
- Change Control — formal CCR/CCB process for production changes.
- Release Management — release to manufacturing governed by gates.
- SDE Archival — product end-of-life with compliant record retention.

9 Real Estate & Property

🏠 Real Estate & Property

Property management platforms, listing marketplaces, investment analytics, smart building systems, and tenant portals.

Real estate businesses build: property listing platforms, tenant management portals, real estate investment analytics tools, smart building management systems, property valuation APIs, and construction project management software.

UC-RE-01 — Build a Property Management SaaS Platform

Actor: PropTech Founder, Developer

Trigger: Startup wants to build a SaaS platform for property managers: tenant management, rent collection, maintenance requests, and reporting.

Goal: Property management SaaS is live with multi-tenant architecture, payment integration, and automated workflows.

Workflow

Multi-Tenant Architecture	Solution: Platform. System: Property Management Platform. Applications: Tenant Portal, Landlord Dashboard, Maintenance Tracking, Rent Collection, Reporting.
Tenant Data Privacy	All tenant PII fields classified. Data masking applied in non-production environments. Retention policies configured per jurisdiction. GDPR/CCPA compliance reports enabled.
Payment Integration	Rent collection application. Threat model for payment flows per UC-SEC-01. PCI-DSS controls documented in change control.
Maintenance Workflow	Maintenance request as Event message: MAINTENANCE_REQUESTED. State message: MAINTENANCE_STATUS. Automated assignment and SLA tracking.
Analytics	Benchmarking Service tracks: occupancy rates, average days-to-let, maintenance SLA completion rate. PM dashboard surfaces KPIs.

Qala Platform Integration

- Privacy Management — PII classification and GDPR/CCPA compliance.
- SEM + Threat Modelling — payment flow security.
- Event/State Messages — maintenance lifecycle as typed messages.
- Benchmarking — property management KPIs.

10 Finance & Investments

💰 Finance & Investments

Fintech applications, investment platforms, trading systems, personal finance tools, and regulatory compliance solutions.

Financial services organisations build: retail banking apps, investment platforms, algorithmic trading systems, personal finance management tools, regulatory reporting services, credit scoring models, and blockchain-based settlement systems. These solutions demand the highest security, audit, and compliance standards.

UC-FIN-01 — Build a Retail Investment Platform

Actor: Fintech Developer, Compliance Officer

Trigger: Fintech startup is building a retail investment platform for stocks, ETFs, and crypto.

Goal: Platform is live with brokerage integration, portfolio management, real-time data, and regulatory compliance.

Workflow

Solution Model	Platform System. Applications: Market Data Feed, Portfolio Management, Trade Execution, Compliance Monitoring, User Onboarding (KYC/AML).
Regulatory Compliance	Compliance requirements (MiFID II, SEC rules, KYC/AML) documented as policy rules in the SEM policy store. Automated compliance checks run on every release.
Security — Elevated	Solution flagged as "Financial — High Security". Enhanced SEM monitoring: all production deployments require SE sign-off. Secrets vault enforced for all credentials. Penetration test required before public launch.
Audit Trail	All trade events, user actions, and configuration changes logged to immutable audit trail. 7-year retention policy applied per regulatory requirement.
Market Data SDE	Market data feed Application deployed in its own SDE with sub-millisecond latency requirements. Hermetic build with frozen dependency set for reproducible performance.
Compliance Report	Monthly compliance report generated from audit log data. Covers: trade audit trail, anomaly detection results, access control review.

Qala Platform Integration

- SEM — elevated security profile; penetration test required pre-launch.
- Secrets Vault — all credentials and API keys managed centrally.
- Immutable Audit Trail — 7-year retention, regulatory audit-ready.
- Compliance Reporting — automated SOC2/MiFID II/KYC compliance dashboards.
- Hermetic Build — reproducible performance for latency-sensitive market data.

UC-FIN-02 — Build an Algorithmic Trading System

Actor: Quant Developer, Risk Manager

Trigger: Investment firm wants to deploy an algorithmic trading strategy.

Goal: Trading algorithm is backtested, validated, deployed to paper trading, then live trading — with full risk controls and audit trail.

Workflow

Research SDE	Quant researcher works in an isolated research SDE. Historical market data versioned in Workspace. Backtesting framework pinned in hermetic environment.
Backtesting Pipeline	Strategy backtested across 10 years of data. Performance metrics (Sharpe ratio, max drawdown, win rate) captured as benchmark records.
Benchmark Gate	Benchmark gate for live deployment: Sharpe ratio ≥ 1.5, max drawdown ≤ 15%. Gate evaluated before paper trading promotion.
Paper Trading	Strategy deployed to paper trading SDE. Live market data consumed; no real capital at risk. Performance monitored for 30 days.
Risk Controls	Risk management Application enforces: position limits, daily loss limits, circuit breakers. These rules are versioned and change-controlled.
Live Deployment	All gates pass. Change Control approved. Strategy deployed to live trading with real-time SEM monitoring for anomalous order patterns.

Qala Platform Integration

- Research SDE — isolated research environment with versioned historical data.
- Benchmark Gate — quantified performance criteria for every promotion.
- Change Control — risk rule changes formally governed.
- SEM — anomalous trading pattern detection in production.
- Immutable Audit Trail — every trade and risk decision logged.

11 Government & Public Sector

🏛 Government & Public Sector

Citizen-facing digital services, case management, permitting, benefits administration, and policy analytics platforms.

Government agencies build: citizen services portals, case management systems, permit and licensing platforms, benefits administration tools, public data portals, and internal workflow automation. They operate under strict compliance, accessibility, and security mandates.

UC-GOV-01 — Modernise a Citizen Services Portal

Actor: Government IT Team, Digital Services Director

Trigger: Government agency wants to replace paper and legacy systems with a modern digital citizen services portal.

Goal: Citizens can apply for services, track status, upload documents, and receive decisions digitally — fully audited and accessible.

Workflow

Discovery	Existing services inventoried. Each service modelled as an Application within the Citizen Services System. Solution type: Service.
Compliance Baseline	Accessibility standard (WCAG 2.1 AA) and security standard (FedRAMP/Cyber Essentials) requirements documented as policy rules in SEM. CI pipeline includes automated accessibility testing.
Identity & Auth	Identity service configured with government identity provider integration (e.g., Gov.uk Verify, Login.gov). MFA mandatory. RBAC defines citizen vs. caseworker roles.
Case Management	Case Application: citizen submits request (event message: CASE_SUBMITTED). Case state tracked through typed state messages. Caseworkers assigned via workflow automation.
Audit Trail	All citizen interactions and caseworker decisions logged to immutable audit trail. FOI (Freedom of Information) request query capability built on audit data.
Staged Rollout	New portal released to 5% of users via canary deployment. Accessibility and usability metrics monitored. Full rollout only after positive confirmation.

Qala Platform Integration

- SEM — government security standard policy enforcement.
- Immutable Audit Trail — FOI-ready citizen interaction logs.
- Identity Service — government IdP integration with mandatory MFA.
- Event/State Messages — case lifecycle as typed messages.
- Canary Deployment — risk-managed public launch.

UC-GOV-02 — Build a Public Open Data Platform

Actor: Government Data Team

Trigger: Agency wants to publish government datasets for public consumption and researcher access.

Goal: Open data portal is live: datasets published, versioned, API-accessible, and usage-tracked.

Workflow

Data Governance	Datasets classified: Public / Restricted / Sensitive. PII fields identified and removed or anonymised before publication. Data pipeline runs anonymisation as a CI stage.
Dataset Versioning	Each dataset version treated as an artefact: timestamped, checksummed, and linked to the publishing release. Consumers can reference specific versions.
API Publication	Data API Application publishes datasets via versioned REST endpoints. OpenAPI spec versioned in CMS. SDK generated automatically.
Usage Analytics	Data Platform tracks: API request counts, most-used datasets, consumer geography. Analytics surfaced on public usage dashboard.
Compliance	Privacy audit report generated quarterly: confirms no PII in published datasets. Anomaly detection alerts if sensitive content detected in new dataset submissions.

Qala Platform Integration

- Privacy Management — PII removal and anonymisation pipelines.
- Artefact Registry — dataset versions with checksums and provenance.
- Workspace/CMS — OpenAPI spec versioning.
- Data Platform — usage analytics and anomaly detection.

12 Coaching, Consulting & Professional Services

🎯 Coaching, Consulting & Professional Services

Coaching platforms, client management systems, methodology tools, knowledge bases, and outcome tracking.

Coaches, consultants, and professional service firms use Qala to build: client relationship management tools, coaching programme delivery platforms, proprietary methodology management systems, outcome tracking apps, knowledge base platforms, and proposal/contract management tools.

UC-COACH-01 — Build a Coaching Programme Delivery Platform

Actor: Coaching Business Owner, Developer

Trigger: Executive coach or coaching firm wants to digitise their programme delivery: sessions, materials, progress tracking, and client communication.

Goal: Coaching platform is live: clients onboard, sessions scheduled, materials delivered, progress tracked, and outcomes reported.

Workflow

Solution Design	Solution type: Platform. System: Coaching Platform. Applications: Client Portal, Session Management, Content Library, Progress Tracker, Outcome Reporting.
Content as Artefacts	Coaching materials (frameworks, worksheets, videos) stored in Workspace/CMS as versioned content artefacts. Version history allows coaches to track methodology evolution.
Programme as Release	Each coaching programme version is treated as a release. New programme version includes: updated content artefacts, revised session flow, new assessment tools. Release gates ensure content completeness before launch.
Client Privacy	Client data (goals, session notes, assessment results) classified as PII. Encryption at rest. Access restricted to assigned coach and client. Data deletion on contract end.
Outcome Benchmarking	Outcome metrics tracked: goal completion rate, NPS, before/after assessment scores. AI Agent identifies which programme elements correlate with highest outcomes.

Qala Platform Integration

- Workspace/CMS — coaching content versioned alongside programme structure.
- Release Management — programme version releases with content completeness gates.
- Privacy Management — client session data PII protection.
- Benchmarking + AI Agent — programme outcome analytics and improvement recommendations.

UC-COACH-02 — Manage a Multi-Consultant Knowledge Platform

Actor: Consulting Firm Knowledge Manager, Consultant

Trigger: Consulting firm wants to capture, manage, and leverage institutional knowledge across consultants and engagements.
Goal: Knowledge platform is live: methodologies documented, reusable frameworks searchable, engagement learnings captured, AI-powered retrieval available.

Workflow

Knowledge as Solutions	Each consulting methodology modelled as a Solution of type Service. Frameworks, templates, and playbooks stored as versioned artefacts in the Workspace/CMS.
Engagement Linking	Each client engagement is a Project in the project management system. Deliverables and learnings linked back to the relevant methodology Solution and versioned.
AI-Powered Retrieval	AI Agent indexes all knowledge artefacts. Consultants query: "What frameworks have we used for supply chain transformation in retail?" AI returns relevant, versioned artefacts with engagement context.
Version Control	When a methodology is updated based on engagement learnings, a new version is created. Change log documents what changed and why. Historical versions preserved for reference.
Access Control	RBAC restricts sensitive client data. Methodologies accessible to all consultants. Client deliverables restricted to engagement team.

Qala Platform Integration

- Workspace/CMS — all knowledge artefacts versioned.
- Project Management — engagements linked to methodology Solutions.
- AI Agent — semantic search and contextual knowledge retrieval.
- RBAC — tiered access for public knowledge vs. confidential client data.

13 Freelancers, Hobbyists & Enthusiasts

🛠️ Freelancers, Hobbyists & Enthusiasts

Personal projects, client work, side hustles, open-source tools, and passion projects — Qala scales down too.

Independent developers, designers, hobbyist builders, and enthusiasts use Qala for: client freelance projects, personal app projects, open-source tool development, hardware/software hobby projects, and independent content platforms. They need powerful tooling without enterprise complexity.

UC-FREELANCE-01 — Manage a Freelance Client Project Portfolio

Actor: Freelance Developer / Designer

Trigger: Freelancer is managing multiple concurrent client projects and wants a single organised environment.

Goal: All client projects are organised in separate SDEs within a personal Solution Factory; delivery is consistent, documented, and auditable.

Workflow

Personal Factory	Freelancer creates a personal Solution Factory. Each client project becomes a separate SDE. Factory-wide policies set (coding standards, security baseline).
Client Onboarding	New client project: SDE provisioned from template, project record created, client workspace configured for file sharing and communication.
Deliverable Tracking	Client deliverables managed as artefacts in each project SDE. Version history provides clear evidence of what was delivered and when.
Billing Evidence	Immutable commit and deployment history provides timestamped evidence of work for billing disputes. Activity reports generated automatically.
Project Handoff	On project completion: SDE archived (with client's approval). All artefacts, documentation, and history packaged. Archive link provided to client.
Template Reuse	Successful project configurations saved as SDE templates. Next similar project spins up in minutes.

Qala Platform Integration

- Personal Solution Factory — all client projects governed together.
- SDE Templates — reusable project starters.
- Artefact Registry — versioned deliverables with timestamps.
- SDE Archival — clean project handoff with full history.
- Audit Trail — activity evidence for billing.

UC-FREELANCE-02 — Build and Publish an Open-Source Tool

Actor: Hobbyist Developer / Open Source Maintainer

Trigger: Developer has built a useful tool and wants to open-source it with proper CI, releases, and documentation.
Goal: Tool is publicly available with hermetic builds, signed releases, automated testing, and versioned documentation.

Workflow

Project SDE	SDE provisioned with the tool's language stack. Hermetic build environment defined: all dependencies pinned.
CI Pipeline	CI configured: every commit triggers hermetic build, unit tests, linting, and SCA. Build must pass before any release.
Signed Releases	Release artefacts (binaries, packages) signed with the maintainer's key. Build attestation generated and published alongside each release.
Changelog	Release notes generated from commit history and linked bug/feature records. Published to Workspace/CMS as versioned documentation.
Community Contributions	Contributor PRs follow UC-DEV-02 code review workflow. CI must pass before maintainer can merge.
Deprecation	When tool reaches end-of-life: final release published, deprecation notice in documentation, SDE archived.

Qala Platform Integration

- Hermetic Build — reproducible builds for all platforms.
- Build Attestation + Signing — tamper-evident release artefacts.
- Release Management — versioned, gated releases.
- Workspace/CMS — versioned documentation and changelogs.

14 Hotel, Hospitality & Travel

🏨 Hotel, Hospitality & Travel

Booking platforms, property management systems, guest experience apps, revenue management tools, and travel marketplaces.

Hospitality businesses build: hotel booking engines, property management systems (PMS), guest experience applications, revenue management platforms, channel managers for OTA distribution, loyalty programme platforms, and travel marketplace solutions.

UC-HOSP-01 — Build a Hotel Property Management System

Actor: Hospitality Tech Team, Hotel Operations Manager
Trigger: Hotel group wants to replace legacy PMS with a modern, integrated system.
Goal: PMS is live: reservations, front desk, housekeeping, maintenance, and reporting all unified.

Workflow

System Design	System: Hotel Operations Platform. Applications: Reservation Management, Front Desk, Housekeeping, Maintenance Requests, Revenue Reporting.
Reservation Events	Check-in and check-out modelled as Event messages. Room availability as State messages. All events published to internal event stream for real-time state synchronisation.
Channel Integration	OTA integrations (Booking.com, Expedia) deployed as separate Applications with typed interface contracts. Rate and availability exported; reservation events imported.
Reliability	Hermetic builds ensure each release is reproducible. Rollback capability critical: a broken front-desk application must be restorable in under 5 minutes.
Revenue Analytics	Benchmarking Service tracks: occupancy rate, RevPAR, ADR, housekeeping completion rate. AI Agent predicts demand and recommends dynamic pricing.
Multi-Property	Solution Factory enrolls each property's SDE. Factory-level benchmarking compares performance across properties.

Qala Platform Integration

- Event/State Architecture — real-time reservation and availability state.
- SDE Rollback — fast recovery for front-desk critical systems.
- Solution Factory — multi-property governance and benchmarking.
- AI Agent — demand prediction and dynamic pricing recommendations.

15 Entertainment, Music & Art

🎵 Entertainment, Music & Art

Music distribution platforms, digital art marketplaces, streaming services, creator tools, ticketing systems, and rights management.

Entertainment, music, and art professionals use Qala to build: independent music distribution platforms, digital art marketplaces (NFT and traditional), streaming content platforms, creator monetisation tools, event ticketing systems, rights and royalty management services, and fan engagement applications.

UC-MUSIC-01 — Build an Independent Music Distribution & Royalty Platform

Actor: Music Tech Developer, Independent Artist / Label

Trigger: Independent label or music tech startup wants to enable artists to distribute music and track royalties.

Goal: Platform live: artists upload music, distribution to streaming services managed, royalties calculated and paid.

Workflow

Platform Design	System: Music Distribution Platform. Applications: Artist Portal, Catalogue Management, Distribution Engine, Royalty Calculator, Payout Management.
Catalogue as Artefacts	Each music track and album treated as a versioned artefact: ISRC, title, contributors, master recording, distribution metadata. Artefact registry stores with full provenance.
DSP Integration	Streaming platform (DSP) integrations deployed as Applications with typed interface contracts. Track export events sent to DSPs; stream count events received.
Royalty Pipeline	Data Platform pipeline processes stream count events. Royalty calculation rules stored as versioned configuration (change-controlled). Royalty statements generated per payout period.
Payout Compliance	Payment flows threat-modelled. Financial audit trail captures every royalty calculation and payout event. Artist can audit their own royalty history.
AI Insights	AI Agent analyses streaming performance: "Track X is gaining traction on playlists in Brazil — recommend targeted promotion".

Qala Platform Integration

- Artefact Registry — music catalogue with ISRC and full provenance.
- Interface Contracts — typed DSP integration event contracts.
- Data Platform — royalty calculation pipelines.
- Change Control — royalty calculation rule changes formally governed.
- Immutable Audit Trail — artist-auditable royalty history.
- AI Agent — streaming trend analysis and promotion recommendations.

UC-ART-01 — Build a Digital Art Marketplace

Actor: Art Marketplace Developer, Digital Artist

Trigger: Startup wants to build a marketplace where digital artists can sell original works and prints.

Goal: Marketplace live: artists list works, buyers purchase, fulfillment managed, artist payments processed.

Workflow

Marketplace Platform	System: Art Marketplace. Applications: Artist Studio (listing management), Buyer Gallery (browsing, purchase), Fulfillment (print-on-demand or digital delivery), Payments.
Artwork Provenance	Each artwork treated as a Solution of type Good. Provenance record: artist identity, creation date, edition details, authenticity certificate. Stored as signed artefact.
Digital Delivery	Digital artwork files stored in Workspace/CMS with access-gated download. Buyer receives signed download token valid for a configurable period.
Print-on-Demand	Physical print orders routed to print partner API. Order events tracked through fulfilment state machine. Shipping integrated per UC-LOGISTICS-01 patterns.
Artist Payments	Payout workflow follows royalty platform patterns: payment events audited, artist dashboard shows earnings, payout scheduled per configurable period.

Qala Platform Integration

- Solution Model — artworks as Solutions of type Good.
- Artefact Registry + Signing — artwork provenance certificates.
- Workspace/CMS — access-gated digital file delivery.
- Data Platform — artist earnings analytics.

16 Academia & Research

🔬 Academia & Research

Experiment management, data pipeline platforms, publication workflows, collaboration tools, and computational research environments.

Academic institutions and research labs use Qala to build and manage: computational experiment pipelines, research data management systems, academic collaboration platforms, publication workflow tools, grant management systems, and reproducible research environments. Reproducibility and provenance are paramount.

UC-RESEARCH-01 — Manage a Reproducible Computational Research Environment

Actor: Research Software Engineer, Principal Investigator

Trigger: Research lab needs to ensure all computational experiments are fully reproducible — anyone in the team can re-run any experiment from any point in time.

Goal: Every experiment runs in a hermetically defined environment; inputs, code, and environment are all versioned; results are reproducible.

Workflow

Research SDE	Each research project provisioned in its own SDE. All software dependencies (libraries, tools, compilers) pinned in hermetic environment definition. No "latest" references.
Dataset Versioning	Research datasets stored in Workspace/CMS with version tags and SHA-256 checksums. Any pipeline run references a specific dataset version.
Experiment as Pipeline	Each experiment is a CI/CD pipeline: input dataset version → code commit → hermetic environment → outputs captured as artefacts with full attestation (what code, what data, what environment produced this result).
Reproducibility Test	A reproducibility gate runs in CI: re-executes the experiment with identical inputs in a fresh hermetic environment and compares outputs to stored reference. Any divergence fails.
Publication Artefacts	Paper submission: all supporting data, code, and environment definitions packaged as a signed artefact bundle. DOI registered. Bundle accessible via permalink for peer review.
Collaboration	Multiple researchers work in separate SDEs within a shared factory. Results artefacts shared via factory-level artefact registry.

Qala Platform Integration

- Hermetic SDE — reproducible environment for all experiments.
- Dataset Versioning — Workspace/CMS with checksums.
- Experiment Pipeline — CI/CD for computational experiments.
- Build Attestation — provenance record for every result.

- Reproducibility Gate — automated re-execution verification.
- Artefact Registry — signed publication bundles.

UC-RESEARCH-02 — Build a Research Data Sharing Platform

Actor: Research Institution, Data Librarian

Trigger: University wants to provide a platform for sharing and discovering research datasets across departments and with external collaborators.

Goal: Data sharing platform live: datasets submitted, reviewed, published, versioned, and discoverable.

Workflow

Data Pipeline	Submission pipeline: researchers upload datasets → automated metadata extraction → data quality checks → PII scan (remove any accidental PII) → review queue.
Review Workflow	Data librarian reviews submissions via portal. Approval follows CCR-style workflow: review, approve, publish.
Dataset Artefacts	Published datasets stored as versioned, checksummed artefacts with: DOI, licence (Creative Commons), contributor list, methodology notes, and related publication links.
Access Control	RBAC: open access datasets available to all; restricted datasets require approval. Access log feeds into audit trail.
Analytics	Usage analytics: most-downloaded datasets, citation tracking, contributor activity. AI Agent identifies high-impact datasets and recommends featured placement.

Qala Platform Integration

- Privacy Management — automated PII detection in submitted datasets.
- Artefact Registry — datasets as versioned, signed artefacts with DOI.
- Change Control (CCR-style) — dataset review and publication workflow.
- RBAC + Audit Trail — access-controlled sharing with full access log.
- AI Agent — impact analytics and discovery recommendations.

17 Consumer Packaged Goods (CPG)

🛒 Consumer Packaged Goods

Product formulation management, supply chain platforms, retail distribution tools, consumer analytics, and sustainability tracking.

CPG companies use Qala to build: product formulation and recipe management systems, packaging design workflows, supply chain visibility platforms, retail distribution management tools, consumer insights platforms, sustainability and carbon tracking systems, and direct-to-consumer e-commerce solutions.

UC-CPG-01 — Manage Product Formulation and Compliance Lifecycle

Actor: Product Development Team, Regulatory Affairs Manager

Trigger: CPG company needs to manage product formulations, ingredient changes, and regulatory compliance across multiple markets.

Goal: All formulations are version-controlled, change-governed, and linked to regulatory approval status per market.

Workflow

Product as Solution	Each product modelled as a Solution of type Good. System: Product Development Platform. Applications: Formulation Management, Regulatory Compliance, Packaging, Market Approvals.
Formulation Versioning	Every formulation is a versioned artefact: ingredient list, quantities, supplier sources, allergen declarations, nutritional profile. Changes require CCR approval.
Regulatory Gate	Release to new market requires regulatory approval gate: market-specific compliance checklist must be completed. Regulatory documents stored as artefacts linked to the release record.
Supplier Changes	Ingredient supplier change triggers CCR: impact assessment required, safety data reviewed, formulation re-validated. All approvals immutably logged.
Recall Readiness	Batch-level traceability: each production run linked to specific formulation version, ingredient batch IDs, and production SDE snapshot. In event of recall, affected batches identified in minutes.

Qala Platform Integration

- Solution Model — CPG products as Solutions of type Good.
- Artefact Registry — formulation versions with full ingredient provenance.
- Change Control — ingredient and supplier changes formally governed.
- Release Management — market approval gates per jurisdiction.
- Immutable Audit Trail — recall-ready batch traceability.

18 Software & Platform Solutions (General)

Software & Platform Solutions

Marketplace platforms, API ecosystems, developer tools, workflow automation, and SaaS products for any vertical.

This section covers general software and platform solution patterns applicable across verticals: two-sided marketplaces, API-first platforms, developer tooling, no-code/low-code platforms, and enterprise workflow automation.

UC-PLATFORM-01 — Build a Two-Sided Marketplace

Actor: Platform Product Team

Trigger: Team is building a marketplace connecting supply-side providers with demand-side consumers.

Goal: Marketplace is live with provider onboarding, consumer discovery, transaction processing, and trust mechanisms.

Workflow

Platform Design	Solution type: Platform. Applications: Provider Portal, Consumer App, Matching Engine, Transaction Processing, Trust & Safety, Analytics.
Provider Onboarding	Provider onboarding flow includes: identity verification, KYC/AML checks (if financial), background checks (if applicable), and policy acceptance. All steps versioned and audited.
Matching Engine	Matching algorithm deployed as Application with defined interface: accepts search query + filters, exports ranked provider list. Algorithm changes require benchmark gate: match quality score must not regress.
Trust & Safety	SEM-powered content moderation: listing content scanned on submission. Fraud detection model monitors transaction patterns. AI Agent flags suspicious accounts.
Seasonal Scaling	Benchmarking ensures platform can handle seasonal traffic spikes. Load tests run before peak periods. Canary deployments used for high-traffic releases.

Qala Platform Integration

- Solution Structure — multi-sided Platform with typed interface contracts.
- Identity + RBAC — provider and consumer role management.
- Benchmark Gate — matching quality gate on algorithm releases.
- SEM + AI Agent — trust and safety monitoring.
- Canary Deployment — safe releases during high-traffic periods.

19 Physical Products & Goods

📦 Physical Products & Goods Businesses

Hardware products, consumer electronics, artisan goods, subscription boxes, and physical product startups.

Physical product businesses use Qala to manage: hardware product development (PCB design, firmware), product lifecycle management, manufacturing coordination, inventory and fulfilment, subscription box curation, and DTC physical product sales.

UC-GOODS-01 — Manage a Hardware + Firmware Product Lifecycle

Actor: Hardware Engineer, Firmware Developer, PM

Trigger: Company is building a consumer IoT device with companion firmware and mobile app.

Goal: Hardware product, firmware, and app are all version-controlled, built hermetically, and released in lockstep.

Workflow

Multi-Solution Model	System: IoT Product Platform. Solutions: Hardware (Good), Firmware (Application), Mobile App (Application), Cloud Backend (Application).
Firmware SDE	Firmware SDE provisioned with embedded toolchain: compiler, linker, flash tool — all pinned in hermetic environment. Cross-compilation reproducible across team.
Hardware-Firmware Dependency	Firmware versions tagged with compatible hardware revision. Interface contract defines firmware-hardware compatibility matrix. Breaking hardware changes require CCR.
OTA Update Pipeline	Over-the-air firmware update pipeline: hermetic build → binary signed → release gated → OTA delivery staged (canary rollout to 1% of devices first).
Mobile-Backend Coordination	Mobile app release and backend API release coordinated in same Release record. API compatibility gate: mobile app must support current and previous API version.
Field Defect Tracking	Customer bug reports from field devices linked to specific firmware version. Defect triage identifies firmware version affected. Fix targeted to that version branch.

Qala Platform Integration

- Multi-Solution Factory — hardware, firmware, app, and backend governed together.
- Hermetic Firmware Build — reproducible cross-compilation.
- Build Signing — OTA binary integrity verification.
- Canary OTA — risk-managed firmware rollout.
- Bug Tracking — field defects linked to specific firmware versions.

20 Education & E-Learning

📚 Education & E-Learning

Online learning platforms, course management, student progress tracking, adaptive learning, and educational content delivery.

UC-EDU-01 — Build an Adaptive Online Learning Platform

Actor: EdTech Developer, Curriculum Designer

Trigger: EdTech company wants to deliver personalised learning paths that adapt to each student's progress.

Goal: Learning platform live with course delivery, progress tracking, adaptive content sequencing, and learner analytics.

Workflow

Platform Design	Solution type: Platform. Applications: Content Library, Learning Path Engine, Student Dashboard, Assessment Engine, Instructor Console, Analytics.
Content as Artefacts	All course content (videos, articles, quizzes, exercises) stored in Workspace/CMS as versioned artefacts. Curriculum updates create new content versions with change log.
Adaptive Algorithm	Learning path recommendation algorithm deployed as Application. Interface: accepts student progress state, exports recommended next content item. AI Agent trains model on completion and assessment data.
Assessment Pipeline	Assessment results stored as state messages per student. Progress events trigger adaptive algorithm. Struggling students flagged for instructor intervention.
Privacy — Students	Student performance data classified as sensitive PII. FERPA/COPPA compliance for under-18 users. Parental consent workflow for minors.
Content Release	New course versions released via release management workflow. Content completeness gate: all modules must have assessment and accessibility review before publish.

Qala Platform Integration

- Workspace/CMS — versioned curriculum content.
- AI Agent — adaptive learning path recommendations.
- Privacy Management — FERPA/COPPA compliance for student data.
- Release Management — content completeness and accessibility gates.

21 Health, Wellness & Medical

Health, Wellness & Medical

Telehealth platforms, patient management, wellness apps, clinical trial tools, and health data analytics.

UC-HEALTH-01 — Build a Telehealth and Patient Management Platform

Actor: Health Tech Developer, Clinical Operations Manager

Trigger: Healthcare provider wants to extend care delivery digitally: appointment booking, teleconsultations, prescription management, and patient records.

Goal: Telehealth platform is live, HIPAA/GDPR-compliant, integrated with EHR, and clinically validated.

Workflow

System Design	System: Telehealth Platform. Applications: Patient Portal, Appointment Scheduling, Video Consultation, Prescription Management, EHR Integration.
HIPAA/GDPR Compliance	All patient data classified as Protected Health Information (PHI). Encryption at rest and in transit enforced. Audit trail captures all PHI access. De-identification pipeline for analytics use.
Security — Clinical	Penetration test required before launch per UC-SEC-03. DAST run against all patient-facing endpoints. Vulnerability SLA: Critical resolved in 4 hours.
EHR Integration	EHR integration Application deploys with HL7 FHIR interface contracts. Events: APPOINTMENT_BOOKED, PRESCRIPTION_ISSUED. State: PATIENT_RECORD.
Clinical Validation	Clinical workflow tested by clinical staff in staging before launch. Sign-off from clinical lead required as an approval gate in the release record.
Reliability	99.99% uptime target. Blue/Green deployments to eliminate downtime during releases. SDE rollback capability for fast incident recovery.

Qala Platform Integration

- Privacy Management — HIPAA PHI classification and de-identification.
- Immutable Audit Trail — PHI access logging for compliance.
- SEM + PenTest — clinical-grade security validation.
- Release Gate — clinical staff sign-off as a mandatory approval.
- Blue/Green Deployment — zero downtime for critical care systems.

22 Industry Use Case Matrix

Summary index of all industry use cases and the primary Qala platform capabilities engaged.

ID	Title	Industry	Key Qala Capabilities
UC-APP-01	Validate a Software App Idea	App Ideas	Prototype SDE, AI Analysis, Solution Mgmt
UC-APP-02	Build and Launch a SaaS Product	App Ideas	CI/CD, Release Gates, SEM, Benchmarking
UC-AI-APP-01	Build an LLM-Powered Application	AI Apps	Model Registry, Eval Pipeline, Canary, Cost Monitoring
UC-AI-APP-02	Manage a Computer Vision Product	AI Apps	Data Platform, ML Registry, Drift Detection
UC-FASHION-01	Launch a DTC Fashion E-Commerce Platform	Fashion	Solution Model, SEM/PCI, Release Mgmt, AI
UC-FASHION-02	Build a Virtual Try-On Feature	Fashion	Prototype SDE, Model Registry, Feature Flags
UC-TECH-01	Operate a Multi-Product Software Company	Technology	Solution Factory, Benchmarking, Shared Registry
UC-TECH-02	Build a Public Developer API Platform	Technology	CMS/OpenAPI, Toolchain SDK Gen, SLA Monitoring
UC-AGRI-01	Build a Precision Farming IoT Platform	Agriculture	Data Platform, Edge SDE, AI Predictions
UC-AGRI-02	Launch a Farm-to-Consumer Marketplace	Agriculture	Platform Design, Event Messages, Analytics
UC-LOGISTICS-01	Build a Real-Time Shipment Tracking Platform	Logistics	Event Architecture, SLA Benchmarking, AI Delay Prediction
UC-LOGISTICS-02	Build a Route Optimisation Service	Logistics	Prototype SDE, Benchmark Gate, AI Improvement
UC-MFG-01	Build a Predictive Maintenance Platform	Manufacturing	Data Platform, Edge SDE, ML Drift Detection
UC-MFG-02	Manage a Digital PLM on Qala	Manufacturing	Solution Model, Version Control, Change Control
UC-RE-01	Build a Property Management SaaS	Real Estate	Privacy Mgmt, SEM, Event Messages, Benchmarking
UC-FIN-01	Build a Retail Investment Platform	Finance	SEM, Audit Trail, Compliance Reporting, Hermetic Build
UC-FIN-02	Build an Algorithmic Trading System	Finance	Research SDE, Benchmark Gate, Change Control, SEM
UC-GOV-01	Modernise a Citizen Services Portal	Government	SEM Policy, Audit Trail, Identity, Canary Deploy
UC-GOV-02	Build a Public Open Data Platform	Government	Privacy Pipeline, Artefact Registry, Analytics

UC-COACH-01	Build a Coaching Programme Platform	Coaching	CMS Versioning, Release Mgmt, Privacy, AI Outcomes
UC-COACH-02	Manage a Multi-Consultant Knowledge Platform	Consulting	CMS, Project Mgmt, AI Retrieval, RBAC
UC-FREELANCE-01	Manage a Freelance Project Portfolio	Freelance	Personal Factory, SDE Templates, Archival, Audit
UC-FREELANCE-02	Build and Publish an Open-Source Tool	Freelance/Hobbyist	Hermetic Build, Signing, Release Mgmt, CMS Docs
UC-HOSP-01	Build a Hotel Property Management System	Hospitality	Event/State Architecture, Rollback, Factory, AI Pricing
UC-MUSIC-01	Build a Music Distribution Platform	Music	Artefact Catalogue, Data Platform, Audit Trail, AI
UC-ART-01	Build a Digital Art Marketplace	Art	Solution Model, Signed Artefacts, CMS, Data Platform
UC-RESEARCH-01	Manage Reproducible Research Environments	Academia	Hermetic SDE, Dataset Versioning, Attestation, Repro Gate
UC-RESEARCH-02	Build a Research Data Sharing Platform	Research	Privacy Pipeline, Artefact Registry, CCR Workflow, AI
UC-CPG-01	Manage Product Formulation Lifecycle	CPG	Solution Model, Artefact Registry, Change Control, Recall
UC-PLATFORM-01	Build a Two-Sided Marketplace	Software/Platform	Platform Design, RBAC, Benchmark Gate, SEM, Canary
UC-GOODS-01	Manage a Hardware + Firmware Lifecycle	Physical Goods	Multi-Solution, Hermetic Firmware, Signing, Canary OTA
UC-EDU-01	Build an Adaptive Learning Platform	Education	CMS Content, AI Adaptive, Privacy (FERPA), Release Gates
UC-HEALTH-01	Build a Telehealth Platform	Health	PHI Privacy, SEM/PenTest, FHIR Interfaces, Blue/Green

End of Document — Qala Solution Factory OS | Extended Workflows & Use Cases — Industry Edition v1.0